

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250272276

Kind Code

A1

Publication Date

August 28, 2025

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GENERATIVE DATA MODELING USING LARGE LANGUAGE MODEL

Abstract

A computer-implemented method may comprise receiving a first user input from a computing device and identifying a plurality of data assets from a catalog of data assets based on the first user input, where each data asset in the catalog of data assets comprises a corresponding entity that comprises data. The method may further comprise causing the identified plurality of data assets to be displayed on the computing device, receiving a first user selection of one or more of the identified plurality of data assets from the computing device, obtaining a plurality of data models based on the one or more of the identified plurality of data assets using a large language model, and causing the plurality of data models to be displayed on the computing device.

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Family ID: 1000007713171

Appl. No.: 18/584307

Filed: February 22, 2024

Publication Classification

Int. Cl.: G06F16/23 (20190101); G06F16/21 (20190101)

U.S. Cl.:

CPC G06F16/2358 (20190101); G06F16/212 (20190101); G06F16/219 (20190101);

Background/Summary

TECHNICAL FIELD

[0001] The present application relates generally to the technical field of computer systems, and, in various embodiments, to systems and methods of generative data modeling using a large language model.

BACKGROUND

[0002] Data models are a foundational element of software development and analytics. They provide a standardized method for defining and formatting database contents consistently across systems, enabling different applications to share the same data. Data modeling is a process used to define and analyze data requirements needed to support processes of an information system. As the diversity and magnitude of data grow, so does the complexity of modeling the data. Current data modeling software tools require users to manually select and configure every element of the data model they are attempting to define, often involving the user navigating through various views and windows to find the proper elements to include in the data model, as well as the proper configuration of the elements in the data model. As a result, current data modeling software tools suffer from an inefficient user interface. Other technical challenges may arise as well.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Some example embodiments of the present disclosure are illustrated by way of example and not limitation in the figures of the accompanying drawings, in which like reference numbers indicate similar elements.

[0004] FIG. 1 is a block diagram illustrating an example generative modeling system.

[0005] FIG. 2 is a flowchart illustrating an example method of implementing generative modeling using a large language model.

[0006] FIG. 3 illustrates an example graphical user interfaces (GUI) in which a plurality of data assets are displayed based on a user input.

[0007] FIG. 4 illustrates an example GUI in which a plurality of data models are displayed based on a user selection of one or more data assets.

[0008] FIG. 5 illustrates an example GUI in which a data model is displayed in response to a user selection of the data model from a plurality of data models.

[0009] FIG. 6 illustrates an example GUI in which the data model displayed in FIG. 5 has been updated based on user input comprising an instruction for editing the data model.

[0010] FIG. 7 is a sequence diagram illustrating an example method of implementing generative modeling using a large language model.

[0011] FIG. 8 is a block diagram illustrating an example catalog.

[0012] FIG. 9 is a block diagram of an example computer system on which methodologies described herein can be executed.

DETAILED DESCRIPTION

[0013] Example methods and systems of generative data modeling using a large language model are disclosed. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of example embodiments. It will be evident, however, to one skilled in the art that the present embodiments can be practiced without these specific details.

[0014] The implementation of the features disclosed herein involves a non-generic, unconventional, and non-routine operation or combination of operations. By applying one or more of the solutions disclosed herein, some technical effects of the system and method of the present disclosure are to implement generative data modeling using a large language model. In some example embodiments, a computer-implemented method may comprise receiving a first user input

from a computing device and identifying a plurality of data assets from a catalog of data assets based on the first user input, where each data asset in the catalog of data assets comprises a corresponding entity that comprises data. The computer-implemented method may further comprise causing the identified plurality of data assets to be displayed on the computing device, receiving a first user selection of one or more of the identified plurality of data assets from the computing device, obtaining a plurality of data models based on the one or more of the identified plurality of data assets using a large language model, and causing the plurality of data models to be displayed on the computing device.

[0015] By using a large language model to generate a plurality of data models based on one or more user-selected data assets, the system and method of the present disclosure improve the user interface of data modeling software tools by significantly reducing the amount of content the user needs to navigate and browse through in defining a data model. As a result, the efficiency of the data modeling process is increased. Other technical effects will be apparent from this disclosure as well.

[0016] The methods or embodiments disclosed herein may be implemented as a computer system having one or more modules (e.g., hardware modules or software modules). Such modules may be executed by one or more hardware processors of the computer system. In some example embodiments, a non-transitory machine-readable storage device can store a set of instructions that, when executed by at least one processor, causes the at least one processor to perform the operations and method steps discussed within the present disclosure.

[0017] The details of one or more variations of the subject matter described herein are set forth in the accompanying drawings and the description below. Other features and benefits of the subject matter described herein will be apparent from the description and drawings, and from the claims.

[0018] FIG. 1 is a block diagram illustrating an example generative modeling system **100**. In some example embodiments, the generative modeling system **100** may comprise any combination of one or more of a user interface component **101**, a generative modeler **102**, a large language model **103**, a catalog of data assets **104**, and a metadata integration layer **105**, and a plurality of data sources **106**. The components shown in FIG. 1 may be configured to communicate with each other via inter-process communication or via one or more network connections.

[0019] One or more of the components of the generative modeling system **100** may be implemented as part of a cloud-based system. For example, one or more of the components of the generative modeling system **100** may be incorporated into an enterprise application platform **120**, providing server-side functionality via a network (e.g., the Internet) to a computing device **110** of a user. The enterprise application platform **120** may comprise web servers and Application Program Interface (API) servers that can be coupled to, and provide web and programmatic interfaces to, application servers. The application servers can be, in turn, coupled to one or more database servers that facilitate access to the data sources **106**. The web servers, API servers, application servers, and database servers can host cross-functional services, which may include relational database modules to provide support services for access to the data sources **106**. Additionally or alternatively, one or more of the components of the generative modeling system **100** may be installed and run on a local on-premise network of the computing device **110** or on the computing device **110** itself.

[0020] In some example embodiments, the generative modeling system **100** may be configured to provide a data modeling software tool that makes the data modeling process easier and more efficient for users of the data modeling software tool. The data modeling software tool may enable the user of the computing device **110** to diagram data flows by interacting with the user interface component **101** via the computing device **110**. For example, when creating a new database structure, the user may provide input to the user interface component **101** to create a diagram of how data will flow into and out of the database. This flow diagram may be used to define the characteristics of the data formats, structures, and database handling functions to efficiently support a particular set of data flow requirements. The data model that results from this process may

provide a framework of relationships between data elements within a database.

[0021] The user interface component **101** may be configured to receive user input from the computing device **110** of the user. The user input may be used by the user interface component **101** to build a data model. For example, the user interface component **101** may provide drag and drop functionality, enabling the user to drag and drop elements into a data model and build connections between elements. The user interface component **101** may additionally or alternatively enable the user to provide other types of user input as well, including, but not limited to, natural language text comprising instructions for creating or editing a data model.

[0022] The generative modeler **102** may be configured to serve as a back-end service that accepts requests from the user interface component **101** and sends the large language model **103** prompts that encapsulate user input for creating a data model and metadata of data assets retrieved from the catalog of data assets **104**. The catalog of data assets **104** may comprise a software application that creates and manages an inventory of an organization's data assets. A data asset may comprise any entity that is comprised of data. Examples of a data asset include, but are not limited to, a table, a database, an output file, and a document. Other types of data assets are also within the scope of the present disclosure. The catalog of data assets **104** may store metadata of data assets. The metadata integration layer **105** may be configured to extract metadata from the data sources **106**. The metadata integration layer **105** may also be configured to keep the metadata stored in the catalog of data assets **104** in sync with changes to the corresponding data assets in the data sources **106**. The large language model, may be used by the generative modeling system **100** in three distinct scenarios: (1) generating initial data models based on metadata from the catalog of data assets **104**, (2) refining a user-selected data model based on user input, and (3) enriching the metadata contained in the catalog of data assets **104**. The features and functions of the components of the generative modeling system **100** will be discussed in further detail below.

[0023] FIG. 2 is a flowchart illustrating an example method **200** of implementing generative modeling using a large language model. The method **200** can be performed by processing logic that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions run on a processing device), or a combination thereof. In one example embodiment, one or more of the operations of the method **200** are performed by the generative modeling system **100** of FIG. 1 or any combination of one or more of its components.

[0024] At operation **202**, the generative modeling system **100** may create or update the catalog of data assets **104**. For example, the metadata integration layer **105** may establish a corresponding network connection between the catalog of data assets **104** and each one of the plurality of data sources **106**, and the catalog of data assets **104** may obtain metadata of the plurality of data assets from the plurality of different data sources **106** using the corresponding network connections and then store the metadata of the plurality of data assets in the catalog of data assets **104**. In some example embodiments, the metadata may comprise one or more of a name for the corresponding data asset, a description for the corresponding data asset, a schema for the corresponding data asset, or a data lineage for the corresponding data asset. However, other types of metadata are also within the scope of the present disclosure.

[0025] The generative modeling system **100** may detect a change in the metadata of one of the data assets in the catalog of data assets **104** using the corresponding network connection with the data source **106** corresponding to the data asset. In response to, or otherwise based on, the detection of the change in the metadata, the generative modeling system **100** may update the catalog of data assets **104** to include the detected change in the metadata of the data asset.

[0026] The generative modeling system **100** may also obtain data lineage of the data assets in the catalog of data assets **104** from the plurality of different data sources **106** using the corresponding network connections. In response to, or otherwise based on, the obtaining of the data lineage of the data assets, the generative modeling system **100** may store the data lineage of the plurality of data assets in the catalog of data assets **104**. Data lineage may provide a record of a data asset

throughout its lifecycle, including source information and any data transformations that have been applied.

[0027] Additionally, the generative modeling system **100** may obtain a corresponding natural language description of the data assets using the large language model **103**. The generative modeling system **100** may obtain the corresponding natural language description by sending the large language model **103**, or another large language model, data of the corresponding data asset (e.g., data included in rows and columns of a table) or metadata of the corresponding data asset. The large language model **103** may generate the corresponding natural language description based on the data or metadata. In response to, or otherwise based on, the obtaining of the corresponding natural language description of the data assets, the generative modeling system **100** may store the corresponding natural language descriptions of the data assets in the catalog of data assets **104**.

[0028] Next, the generative modeling system **100** may, at operation **204**, receive a first user input from the computing device **110**. In some example embodiments, the first user input may comprise one or more terms entered by a user of the computing device **110** into a search field displayed on the computing device **110**. However, other types of user input are also within the scope of the present disclosure.

[0029] The generative modeling system **100** may then identify a plurality of data assets from the catalog of data assets **104** based on the first user input, at operation **206**. Each data asset in the catalog of data assets **104** may comprise a corresponding entity that comprises data. For example, one or more of the plurality of data assets may comprises a table of data. However, other types of data assets are also within the scope of the present disclosure. In some example embodiments, the identifying of the plurality of data assets may comprise searching the catalog of data assets **104** for the first user input, and, for each data asset in the plurality of data assets, determining that data (e.g., data in cells of a table) or metadata of the data asset matches the first user input based on the searching of the catalog of data assets **104**. The identifying of the plurality of data assets may be based on the determination that the data or metadata of the plurality of data assets matches the first user input. In some example embodiments, the identifying of the plurality of data assets may comprise comparing the first user input to the metadata of the plurality of data assets stored in the catalog of data assets. The identifying of the plurality of data assets may additionally or alternatively be based on the data lineage of the plurality of data assets stored in the catalog of data assets **104** or the corresponding natural language descriptions of the plurality of data assets stored in the catalog of data assets **104**.

[0030] At operation **208**, the generative modeling system **100** may cause the identified plurality of data assets to be displayed on the computing device **110**. The displaying of the identified plurality of data assets may comprise displaying representations of the data assets. For example, the generative modeling system **100** may display a box containing a name of a table to represent the table as opposed to displaying the contents of the table itself.

[0031] Next, the generative modeling system **100** may, at operation **210**, receive a first user selection of one or more of the identified plurality of data assets from the computing device **110**. FIG. 3 illustrates an example GUI **300** in which a plurality of data assets **320** are displayed based on a user input. In the example shown in FIG. 3, the plurality of data assets **320** comprises data asset **320-1** corresponding to a table of sales orders, data asset **320-2** corresponding to a table of bike products, data asset **320-3** corresponding to a table of sales managers, data asset **320-4** corresponding to a table of bike parts, data asset **320-5** corresponding to a table of shippers, data asset **320-6** corresponding to a table of customers, data asset **320-7** corresponding to a table of costs, and data asset **320-8** corresponding to a table of factories. The plurality of data assets **320** may be displayed in response to the user of the computing device **110** submitting a first user input, such as by the user entering the terms “bike sales” into a search field **310** displayed on the computing device **110**. The GUI **300** may be configured to receive a first user selection of one or more of the identified plurality of data assets **320** from the user. For example, the user may click

on, or otherwise select, data assets **320-1**, **320-2**, and **320-3**, and then select a user interface element **330** configured to submit these selected data assets as the first user selection for use in generating data models.

[0032] Referring back to FIG. 2, at operation **212**, the generative modeling system **100** may obtain a plurality of data models based on the one or more of the identified plurality of data assets using the large language model **103**. The large language model **103** may comprise an artificial intelligence algorithm that uses deep learning techniques and massively large data sets to understand, summarize, and generate data models. The plurality of data models may comprise the one or more data assets selected by the user.

[0033] In some example embodiments, the obtaining of the plurality of data models may comprise retrieving metadata of the one or more of the identified plurality of data assets from the catalog of data assets **104** based on the first user selection, sending a first request to the large language model **103**, where the first request comprises the retrieved metadata and is configured to prompt the large language model **103** to generate the plurality of data models based on the retrieved metadata, and receiving the plurality of data models from the large language model **103**. The retrieved metadata may comprise one or more of a name for the corresponding data asset, a description for the corresponding data asset, a schema for the corresponding data asset, or a data lineage for the corresponding data asset. However, other types of metadata are also within the scope of the present disclosure.

[0034] The obtaining of the plurality of data models may further comprise determining that one of the plurality of data models does not satisfy a format or grammar rule. For example, the large language model **103** may return the plurality of data models to the generative modeler **102** in code format, such as shown in the following example below:

```
TABLE-US-00001 [ {      "id" : "id0",      "operator" : "join",      "parameters" : {
    "left" : "SalesOrders.PRODUCT_ID",      "right" : "BikeProducts.ID",      "type" :
"inner"      } }, {      "id" : "id1",      "operator" : "filter",      "parameters" : {
    "operator" : "equal",      "property" : "STATUS",      "value" : "FINISHED"      }
}. {      "id" : "id2",      "operator" : "join",      "parameters" : {      "left" :
"id1.SALES_MANAGER_ID",      "right" : "SalesManagers.ID",      "type" : "inner"
      } }, {      "id" : "id3",      "operator" : "output",      "parameters" : {      "name" :
"SalesReport"      } } ]
```

[0035] The generative modeler **102** may access a set of format and grammar rules to determine whether the plurality of data models satisfy the set of format and grammar rules. The set of format and grammar rules may comprise at least one format rule specifying how to format the code representing a data model (e.g., conventions for the style of the code), as well as at least one grammar rule specifying how to write code statements that are valid for a programming language (e.g., rules that specify how characters and words may be put one after the other to form valid statements) in order to guarantee that the generated code is runnable on the generative modeling system **100**. The set of format and grammar rules may be stored by the generative modeler **102** based on specifications input by a user of the generative modeling system **100**. The generative modeler **102** may be configured to detect format and grammar errors it detects in the data models it receives from the large language model **103**. In some example embodiments, the generative modeler **102** may send an indication that the one of the plurality of data models does not satisfy the format rule to the large language model **103**. The indication may include details identifying the format rule and the portion of the code representing the data model that does not comply with the format rule. The large language model **103** may then generate a corrected version of the data model using the details included in the indication, and then return the corrected version of the data model. The generative modeler **102** may then receive a corrected version of the data model.

[0036] The generative modeling system **100** may, at operation **214**, cause the plurality of data models to be displayed on the computing device. The generative modeling system **100** may process

the code representing the data models returned by the large language model **103** and render visual representations of the data models based on the processing of the code. In some example embodiments, the generative modeling system **100** may display the data models in a carousel, such as one data model displayed on the screen of the computing device **110** at a time, while displaying one or more user interface elements configured to navigate from one data model to another upon selection by the user. Alternatively, the generative modeling system **100** may also display all of the data models concurrently on the same page or within the same view.

[0037] The generative modeling system **100** may receive a second user selection of one of the plurality of data models from the computing device **110**, at operation **216**. FIG. **4** illustrates an example GUI **400** in which a plurality of data models **420** are displayed based on a user selection of one or more data assets. In the example GUI **400**, the data models **420** are displayed one at a time within a window **410**, and the user may navigate from one data model **420** to another by selecting user interface elements **412** or **414**. In the example shown in FIG. **4**, the data model **420** being displayed comprises data assets **320-1**, **320-2**, and **330-3**, as well as other elements, such as a filter **424** and an output file **422**. The GUI **400** may be configured to receive the second user selection of one of the data models **420** from the user, thereby enabling the user to select one of the plurality of data models **420** for further modeling. For example, the user may click on, or otherwise select, a user interface element **430** configured to submit the data model **420** currently being displayed in the window **410** as the second user selection.

[0038] At operation **218**, the generative modeling system **100** may cause the selected data model to be displayed on the computing device **110** based on the second user selection. The generative modeling system **100** may also cause the display of one or more user interface elements configured to enable the user to provide user input for editing the selected data model.

[0039] The generative modeling system **100** may, at operation **220**, receive a second user input from the computing device **110**. In some example embodiments, the second user input may comprise one or more instructions for editing the selected data model displayed on the computing device **110**. The second user input may comprise a natural language prompt, and the data model may be updated based on the natural language prompt using the large language model **103** or another large language model.

[0040] FIG. **5** illustrates an example GUI **500** in which the data model **420** is displayed in response to a user selection of the data model **420** from the plurality of data models **420**. In FIG. **5**, the GUI **500** comprises a conversation panel **510** that displays the communication between the user of the computing device **110** and a chatbot of the generative modeling system **100**. In the example shown in FIG. **5**, the user has submitted the second user input in the form of an instruction **512** for editing the selected data model **420**. The instruction **512** may be entered by the user via a text field **520**, and, upon submission by the user, may be displayed within the conversation panel **510**.

[0041] Referring back to FIG. **2**, at operation **222**, the generative modeling system **100** may update the selected data model **420** displayed on the computing device **110** based on the one or more instructions for editing the selected data model **420**. FIG. **6** illustrates an example GUI **600** in which the data model **420** displayed in FIG. **5** has been updated based on user input comprising an instruction for editing the data model **420**. For example, the instruction provided by the user in FIG. **5** was “PLEASE SHOW ONLY THE TOP **10** SALES MANAGERS IN THE REPORT.” Based on this instruction, the generative modeling system **100** has generated an updated version of the data model **420'** that has added a filter **626** in accordance with the instruction. The generative modeling system **100** may display a confirmation **614** in the conversation panel **510** to indicate that the instruction for editing the data model has been executed. The user may provide multiple rounds of instructions for editing the data model **420** or the updated version of the data model **420'**. When the user is satisfied with the data model, the user may provide input instructing the generative modeling system **100** to store the data model for subsequent use.

[0042] It is contemplated that any of the other features described within the present disclosure can

be incorporated into the method **200**.

[0043] FIG. 7 is a sequence diagram illustrating an example method **700** of implementing generative modeling using a large language model. The method **700** can be performed by processing logic that can comprise hardware (e.g., circuitry, dedicated logic, programmable logic, microcode, etc.), software (e.g., instructions run on a processing device), or a combination thereof. [0044] At operation **701**, a user of the computing device **110** may provide a first user input via the computing device **110**, and the computing device **110** may send the first user input to the catalog of data assets **104**. The first user input may comprise one or more terms entered by the user into a search field displayed on the computing device **110**. At operation **702**, the catalog of data assets **104** identifies a plurality of data assets from the catalog of data assets **104** based on the first user input, and then sends the identified plurality of data assets back to the computing device **110** for display on the computing device **110**. Each data asset in the catalog of data assets **104** may comprise a corresponding entity that comprises data. In some example embodiments, the catalog of data assets **104** may identify the plurality of data assets by searching the catalog of data assets **104** for the first user input, and then determining that data or metadata of the plurality of data assets matches the first user input based on the searching of the catalog of data assets **104**, where the identifying of the plurality of data assets is based on the determination that the data or metadata of the plurality of data assets matches the first user input.

[0045] At operation **703**, the user interface component **101** may receive a first user selection of one or more of the identified plurality of data assets from the computing device **110**. In response to receiving the first user selection, the user interface component **101** may, at operation **704**, send a request to the generative modeler **102** to start a data modeling process. This request may comprise an identification of the one or more data assets selected by the user. In response to receiving the request to start the modeling process, the generative modeler **102** may send a request to the catalog of data assets **104** to get metadata of the one or more selected data assets based on the first user selection, at operation **705**. At operation **706**, the catalog of data assets **104** may search for metadata stored in association with the one or more selected data assets, and then return the associated metadata to the generative modeler **102**. The metadata may comprise one or more of a name for the corresponding data asset, a description for the corresponding data asset, a schema for the corresponding data asset, or a data lineage for the corresponding data asset. Other types of metadata are also within the scope of the present disclosure.

[0046] At operation **707**, the generative modeler **102** may send a request to the large language model **103**. The request may comprise the metadata and be configured to prompt the large language model **103** to generate a plurality of data models based on the metadata. The large language model **103** may then, at operation **708**, generate the plurality of data models using the metadata and return the generated plurality of data models back to the generative modeler **102**. The generative modeler **102** may optionally verify whether the plurality of data models satisfy a set of format and grammar rules. For example, the set of format and grammar rules may comprise at least one format rule specifying how to format the code representing a data model (e.g., conventions for the style of the code), as well as at least one grammar rule specifying how to write code statements that are valid for a programming language (e.g., rules that specify how characters and words may be put one after the other to form valid statements) in order to guarantee that the generated code is runnable on the generative modeling system **100**. The set of format and grammar rules may be stored by the generative modeler **102** based on specifications input by a user of the generative modeling system **100**. The generative modeler **102** may be configured to detect format and grammar errors it detects in the data models it receives from the large language model **103**. If one of the data models does not satisfy a particular formal or grammar rule, then the generative modeler **102** may, at operation **709**, send an indication of this error in the data model to the large language model **103**. The large language model **103** may then use this indication to generate a corrected version of the data model, and return the corrected version of the data model to the generative modeler, at operation **710**.

[0047] The generative modeler **102** may, at operation **711**, send the data models to the user interface component **101**, which may then cause the data models to be displayed on the computing device **110**, at operation **712**. At operation **713**, the user interface component **101** may receive a second user selection of one of the plurality of data models from the computing device **110**. The user interface component **101** may then cause the selected data model to be displayed on the computing device **110**, at operation **714**, based on the second user selection and display one or more user interface elements configured to enable the user to fine-tune the selected data model. [0048] The user may engage in a looping fine-tuning process in which the user provides, via the computing device **110**, a user input comprising one or more instructions for editing the selected data model to the user interface component **101**, at operation **715**, and the user interface component **101** sends a corresponding request to update the selected data model to the generative modeler **102**, at operation **716**. At operation **717**, the generative modeler **102** may send a prompt to the large language model **103** to update the selected model. This prompt may comprise an indication of the one or more instructions for editing the selected data model provided by the user. The large language model **103** may then, at operation **718**, update the selected data model based on the one or more instructions for editing the selected data model and return the updated model to the generative modeler **102**, which may send the updated model to the user interface component **101**, at operation **719**. At operation **720**, the user interface component **101** may then cause the updated model to be displayed on the computing device **110** for the user to decide whether to adopt the current version of the data model or continue refining the data model. This cycle of fine-tuning can be repeated multiple times until the user attains an ideal data model.

[0049] It is contemplated that any of the other features described within the present disclosure can be incorporated into the method **700**.

[0050] FIG. **8** is a block diagram illustrating an example catalog of data assets **104**. In some example embodiments, the catalog of data assets **104** may comprise a metadata repository **810**, a change capture component **820**, a metadata enrichment component **830**, and a lineage analysis component **840**. The change capture component **820** may be configured to manage extraction of metadata from the data sources **106** via the metadata integration layer **105**. The change capture component **820** may track metadata alterations in the data sources **106**, and then update the metadata repository based on the tracked alterations.

[0051] The metadata enrichment component **830** may be configured to send a request to the large language model **103** to generate a corresponding natural language description of each one of the plurality of data assets in the catalog of data assets **104**. The request may include data or metadata of each data asset. The metadata enrichment component **830** may then store the natural language descriptions in association with their corresponding data assets in the catalog of data assets **104**. The metadata enrichment component **830** may also be configured to automatically generate business tags and glossary terms for each data asset based on the data or metadata of the data asset. The metadata enrichment component **830** may send a request to the large language model **103** to generate one or more business tags or glossary terms for a data asset based on the data or metadata of the data asset. The metadata enrichment component **830** may store the generated business tags and glossary terms in association with their corresponding data assets in the catalog of data assets **104**.

[0052] The lineage analysis component **840** may be configured to fetch lineage information from the data sources **106** via the metadata integration layer **105**. The data gathered by the change capture component **820**, the metadata enrichment component **830**, and the lineage analysis component **840** may be stored in the metadata repository **810** for subsequent transmission to the large language model **103** as metadata of the corresponding data asset(s) for use in generating the data models.

[0053] In view of the disclosure above, various examples are set forth below. It should be noted that one or more features of an example, taken in isolation or combination, should be considered

within the disclosure of this application.

[0054] Example 1 includes a computer-implemented method performed by a computer system having a memory and at least one hardware processor, the computer-implemented method comprising: receiving a first user input from a computing device; identifying a plurality of data assets from a catalog of data assets based on the first user input, each data asset in the catalog of data assets comprising a corresponding entity that comprises data; causing the identified plurality of data assets to be displayed on the computing device; receiving a first user selection of one or more of the identified plurality of data assets from the computing device; obtaining a plurality of data models based on the one or more of the identified plurality of data assets using a large language model; and causing the plurality of data models to be displayed on the computing device.

[0055] Example 2 includes the computer-implemented method of example 1, wherein the first user input comprises one or more terms entered by a user of the computing device into a search field displayed on the computing device.

[0056] Example 3 includes the computer-implemented method of example 1 or example 2, wherein one or more of the plurality of data assets comprises a table of data.

[0057] Example 4 includes the computer-implemented method of any one of examples 1 to 3, wherein the identifying of the plurality of data assets comprises: searching the catalog of data assets for the first user input; and for each data asset in the plurality of data assets, determining that data or metadata of the data asset matches the first user input based on the searching of the catalog of data assets, wherein the identifying of the plurality of data assets is based on the determining that the data or metadata of the plurality of data assets matches the first user input.

[0058] Example 5 includes the computer-implemented method of any one of examples 1 to 4, wherein the plurality of data models comprises the identified plurality of data assets.

[0059] Example 6 includes the computer-implemented method of any one of examples 1 to 5, wherein the obtaining of the plurality of data models comprises: retrieving metadata of the one or more of the identified plurality of data assets from the catalog of data assets based on the first user selection; sending a first request to the large language model, the first request comprising the retrieved metadata and being configured to prompt the large language model to generate the plurality of data models based on the retrieved metadata; and receiving the plurality of data models from the large language model.

[0060] Example 7 includes the computer-implemented method of any one of examples 1 to 6, wherein the obtaining of the plurality of data models further comprises: determining that one of the plurality of data models does not satisfy a rule, the rule comprising a format rule or a grammar rule; sending an indication that the one of the plurality of data models does not satisfy the rule to the large language model; and receiving a corrected version of the one of the plurality of data models based on the sending of the indication.

[0061] Example 8 includes the computer-implemented method of any one of examples 1 to 7, wherein the retrieved metadata comprises one or more of a name for the corresponding data asset, a description for the corresponding data asset, a schema for the corresponding data asset, or a data lineage for the corresponding data asset.

[0062] Example 9 includes the computer-implemented method of any one of examples 1 to 8, further comprising: establishing a corresponding network connection with each one of a plurality of data sources; obtaining metadata of the plurality of data assets from the plurality of different data sources using the corresponding network connections; and storing the metadata of the plurality of data assets in the catalog of data assets, wherein the identifying of the plurality of data assets comprises comparing the first user input to the metadata of the plurality of data assets stored in the catalog of data assets.

[0063] Example 10 includes the computer-implemented method of any one of examples 1 to 9, further comprising: detecting a change in the metadata of one of the data assets in the plurality of data assets; and updating the catalog of data assets to include the detected change in the metadata

of the one of the data assets in the plurality of data assets.

[0064] Example 11 includes the computer-implemented method of any one of examples 1 to 10, further comprising: obtaining data lineage of the plurality of data assets from the plurality of different data sources using the corresponding network connections; and storing the data lineage of the plurality of data assets in the catalog of data assets, wherein the identifying of the plurality of data assets is further based on the data lineage of the plurality of data assets stored in the catalog of data assets.

[0065] Example 12 includes the computer-implemented method of any one of examples 1 to 11, further comprising: obtaining a corresponding natural language description of each one of the plurality of data assets using the large language model or another large language model; and storing the corresponding natural language descriptions of the plurality of data assets in the catalog of data assets, wherein the identifying of the plurality of data assets is further based on the corresponding natural language descriptions of the plurality of data assets stored in the catalog of data assets.

[0066] Example 13 includes the computer-implemented method of any one of examples 1 to 12, further comprising: receiving a second user selection of one of the plurality of data models from the computing device; and causing the one of the plurality of data models to be displayed on the computing device based on the second user selection.

[0067] Example 14 includes the computer-implemented method of any one of examples 1 to 13, further comprising: receiving a second user input from the computing device, the second user input comprising one or more instructions for editing the one of the plurality of data models displayed on the computing device; and updating the one of the plurality of data models displayed on the computing device based on the one or more instructions for editing the one of the plurality of data models.

[0068] Example 15 includes the computer-implemented method of any one of examples 1 to 14, wherein the second user input comprises a natural language prompt, and the one of the plurality of data models is updated based on the natural language prompt using the large language model or another large language model.

[0069] Example 16 includes a system comprising: at least one processor; and a non-transitory computer-readable medium storing executable instructions that, when executed, cause the at least one processor to perform the method of any one of examples 1 to 15.

[0070] Example 17 includes a non-transitory machine-readable storage medium, tangibly embodying a set of instructions that, when executed by at least one processor, causes the at least one processor to perform the method of any one of examples 1 to 15.

[0071] Example 18 includes a machine-readable medium carrying a set of instructions that, when executed by at least one processor, causes the at least one processor to carry out the method of any one of examples 1 to 15.

[0072] Certain embodiments are described herein as including logic or a number of components, modules, or mechanisms. Modules may constitute either software modules (e.g., code embodied on a machine-readable medium or in a transmission signal) or hardware modules. A hardware module is a tangible unit capable of performing certain operations and may be configured or arranged in a certain manner. In example embodiments, one or more computer systems (e.g., a standalone, client, or server computer system) or one or more hardware modules of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware module that operates to perform certain operations as described herein.

[0073] The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented modules that operate to perform one or more operations or functions. The modules referred to herein may, in some example

embodiments, comprise processor-implemented modules.

[0074] Similarly, the methods described herein may be at least partially processor-implemented. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented modules. The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processor or processors may be located in a single location (e.g., within a home environment, an office environment or as a server farm), while in other embodiments the processors may be distributed across a number of locations.

[0075] The one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), these operations being accessible via a network (e.g., the network **114** of FIG. **1**) and via one or more appropriate interfaces (e.g., APIs).

[0076] Example embodiments may be implemented in digital electronic circuitry, or in computer hardware, firmware, software, or in combinations of them. Example embodiments may be implemented using a computer program product, e.g., a computer program tangibly embodied in an information carrier, e.g., in a machine-readable medium for execution by, or to control the operation of, data processing apparatus, e.g., a programmable processor, a computer, or multiple computers.

[0077] A computer program can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, subroutine, or other unit suitable for use in a computing environment. A computer program can be deployed to be executed on one computer or on multiple computers at one site or distributed across multiple sites and interconnected by a communication network.

[0078] In example embodiments, operations may be performed by one or more programmable processors executing a computer program to perform functions by operating on input data and generating output. Method operations can also be performed by, and apparatus of example embodiments may be implemented as, special purpose logic circuitry (e.g., a FPGA or an ASIC).

[0079] FIG. **9** is a block diagram of a machine in the example form of a computer system **900** within which instructions **924** for causing the machine to perform any one or more of the methodologies discussed herein may be executed. In alternative embodiments, the machine operates as a standalone device or may be connected (e.g., networked) to other machines. In a networked deployment, the machine may operate in the capacity of a server or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine may be a personal computer (PC), a tablet PC, a set-top box (STB), a Personal Digital Assistant (PDA), a cellular telephone, a web appliance, a network router, switch or bridge, or any machine capable of executing instructions (sequential or otherwise) that specify actions to be taken by that machine. Further, while only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

[0080] The example computer system **900** includes a processor **902** (e.g., a central processing unit (CPU), a graphics processing unit (GPU) or both), a main memory **904**, and a static memory **906**, which communicate with each other via a bus **908**. The computer system **900** may further include a graphics or video display unit **910** (e.g., a liquid crystal display (LCD) or a cathode ray tube (CRT)). The computer system **900** also includes an alphanumeric input device **912** (e.g., a keyboard), a user interface (UI) navigation (or cursor control) device **914** (e.g., a mouse), a storage unit (e.g., a disk drive unit) **916**, an audio or signal generation device **918** (e.g., a speaker), and a network interface device **920**.

[0081] The storage unit **916** includes a machine-readable medium **922** on which is stored one or more sets of data structures and instructions **924** (e.g., software) embodying or utilized by any one or more of the methodologies or functions described herein. The instructions **924** may also reside, completely or at least partially, within the main memory **904** and/or within the processor **902** during execution thereof by the computer system **900**, the main memory **904** and the processor **902** also constituting machine-readable media. The instructions **924** may also reside, completely or at least partially, within the static memory **906**.

[0082] While the machine-readable medium **922** is shown in an example embodiment to be a single medium, the term “machine-readable medium” may include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more instructions **924** or data structures. The term “machine-readable medium” shall also be taken to include any tangible medium that is capable of storing, encoding or carrying instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies of the present embodiments, or that is capable of storing, encoding or carrying data structures utilized by or associated with such instructions. The term “machine-readable medium” shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media. Specific examples of machine-readable media include non-volatile memory, including by way of example semiconductor memory devices (e.g., Erasable Programmable Read-Only Memory (EPROM), Electrically Erasable Programmable Read-Only Memory (EEPROM), and flash memory devices); magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and compact disc-read-only memory (CD-ROM) and digital versatile disc (or digital video disc) read-only memory (DVD-ROM) disks.

[0083] The instructions **924** may further be transmitted or received over a communications network **926** using a transmission medium. The instructions **924** may be transmitted using the network interface device **920** and any one of a number of well-known transfer protocols (e.g., HTTP). Examples of communication networks include a LAN, a WAN, the Internet, mobile telephone networks, POTS networks, and wireless data networks (e.g., WiFi and WiMAX networks). The term “transmission medium” shall be taken to include any intangible medium capable of storing, encoding, or carrying instructions for execution by the machine, and includes digital or analog communications signals or other intangible media to facilitate communication of such software.

[0084] This detailed description is merely intended to teach a person of skill in the art further details for practicing certain aspects of the present teachings and is not intended to limit the scope of the claims. Therefore, combinations of features disclosed above in the detailed description may not be necessary to practice the teachings in the broadest sense, and are instead taught merely to describe particularly representative examples of the present teachings.

[0085] Unless specifically stated otherwise, discussions utilizing terms such as “processing” or “computing” or “calculating” or “determining” or “displaying” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

[0086] Although an embodiment has been described with reference to specific example embodiments, it will be evident that various modifications and changes may be made to these embodiments without departing from the broader spirit and scope of the present disclosure. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. The accompanying drawings that form a part hereof show, by way of illustration, and not of limitation, specific embodiments in which the subject matter may be practiced. The embodiments illustrated are described in sufficient detail to enable those skilled in the art to practice the teachings disclosed herein. Other embodiments may be utilized and derived therefrom,

such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. This Detailed Description, therefore, is not to be taken in a limiting sense, and the scope of various embodiments is defined only by the appended claims, along with the full range of equivalents to which such claims are entitled.

[0087] The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in a single embodiment for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separate embodiment.

Claims

1. A computer-implemented method comprising: receiving a first user input from a computing device; identifying a plurality of data assets from a catalog of data assets based on the first user input, each data asset in the catalog of data assets comprising a corresponding entity that comprises data; causing the identified plurality of data assets to be displayed on the computing device; receiving a first user selection of one or more of the identified plurality of data assets from the computing device; retrieving metadata of the one or more of the identified plurality of data assets from the catalog of data assets based on the first user selection; sending a first request to a large language model, the first request comprising the retrieved metadata and being configured to prompt the large language model to generate a plurality of data models based on the retrieved metadata; receiving the plurality of data models from the large language model; determining that one of the plurality of data models does not satisfy a rule, the rule comprising a format rule or a grammar rule; sending an indication that the one of the plurality of data models does not satisfy the rule to the large language model; receiving a corrected version of the one of the plurality of data models based on the sending of the indication; and causing the plurality of data models to be displayed on the computing device.
2. The computer-implemented method of claim 1, wherein the first user input comprises one or more terms entered by a user of the computing device into a search field displayed on the computing device.
3. The computer-implemented method of claim 1, wherein one or more of the plurality of data assets comprises a table of data.
4. The computer-implemented method of claim 1, wherein the identifying of the plurality of data assets comprises: searching the catalog of data assets for the first user input; and for each data asset in the plurality of data assets, determining that data or metadata of the data asset matches the first user input based on the searching of the catalog of data assets, wherein the identifying of the plurality of data assets is based on the determining that the data or metadata of the plurality of data assets matches the first user input.
5. The computer-implemented method of claim 1, wherein the plurality of data models comprises the identified plurality of data assets.
6. (canceled)
7. (canceled)
8. The computer-implemented method of claim 1, wherein the retrieved metadata comprises one or more of a name for the corresponding data asset, a description for the corresponding data asset, a schema for the corresponding data asset, or a data lineage for the corresponding data asset.

9. The computer-implemented method of claim 1, further comprising: establishing a corresponding network connection with each one of a plurality of data sources; obtaining metadata of the plurality of data assets from the plurality of data sources using the corresponding network connections; and storing the metadata of the plurality of data assets in the catalog of data assets, wherein the identifying of the plurality of data assets comprises comparing the first user input to the metadata of the plurality of data assets stored in the catalog of data assets.

10. The computer-implemented method of claim 9, further comprising: detecting a change in the metadata of one of the data assets in the plurality of data assets; and updating the catalog of data assets to include the detected change in the metadata of the one of the data assets in the plurality of data assets.

11. The computer-implemented method of claim 9, further comprising: obtaining data lineage of the plurality of data assets from the plurality of data sources using the corresponding network connections; and storing the data lineage of the plurality of data assets in the catalog of data assets, wherein the identifying of the plurality of data assets is further based on the data lineage of the plurality of data assets stored in the catalog of data assets.

12. The computer-implemented method of claim 9, further comprising: obtaining a corresponding natural language description of each one of the plurality of data assets using the large language model or another large language model; and storing the corresponding natural language descriptions of the plurality of data assets in the catalog of data assets, wherein the identifying of the plurality of data assets is further based on the corresponding natural language descriptions of the plurality of data assets stored in the catalog of data assets.

13. The computer-implemented method of claim 1, further comprising: receiving a second user selection of one of the plurality of data models from the computing device; and causing the one of the plurality of data models to be displayed on the computing device based on the second user selection.

14. The computer-implemented method of claim 13, further comprising: receiving a second user input from the computing device, the second user input comprising one or more instructions for editing the one of the plurality of data models displayed on the computing device; and updating the one of the plurality of data models displayed on the computing device based on the one or more instructions for editing the one of the plurality of data models.

15. The computer-implemented method of claim 14, wherein the second user input comprises a natural language prompt, and the one of the plurality of data models is updated based on the natural language prompt using the large language model or another large language model.

16. A system comprising: at least one hardware processor; and a non-transitory computer-readable medium storing executable instructions that, when executed, cause the at least one hardware processor to perform computer operations comprising: receiving a first user input from a computing device; identifying a plurality of data assets from a catalog of data assets based on the first user input, each data asset in the catalog of data assets comprising a corresponding entity that comprises data; causing the identified plurality of data assets to be displayed on the computing device; receiving a first user selection of one or more of the identified plurality of data assets from the computing device; retrieving metadata of the one or more of the identified plurality of data assets from the catalog of data assets based on the first user selection; sending a first request to a large language model, the first request comprising the retrieved metadata and being configured to prompt the large language model to generate a plurality of data models based on the retrieved metadata; receiving the plurality of data models from the large language model; determining that one of the plurality of data models does not satisfy a rule, the rule comprising a format rule or a grammar rule; sending an indication that the one of the plurality of data models does not satisfy the rule to the large language model; receiving a corrected version of the one of the plurality of data models based on the sending of the indication; and causing the plurality of data models to be displayed on the computing device.

- 17.** The system of claim 16, wherein the first user input comprises one or more terms entered by a user of the computing device into a search field displayed on the computing device.
- 18.** The system of claim 16, wherein one or more of the plurality of data assets comprises a table of data.
- 19.** The system of claim 16, wherein the identifying of the plurality of data assets comprises: searching the catalog of data assets for the first user input; and for each data asset in the plurality of data assets, determining that data or metadata of the data asset matches the first user input based on the searching of the catalog of data assets, wherein the identifying of the plurality of data assets is based on the determining that the data or metadata of the plurality of data assets matches the first user input.
- 20.** A non-transitory machine-readable storage medium tangibly embodying a set of instructions that, when executed by at least one hardware processor, causes the at least one hardware processor to perform computer operations comprising: receiving a first user input from a computing device; identifying a plurality of data assets from a catalog of data assets based on the first user input, each data asset in the catalog of data assets comprising a corresponding entity that comprises data; causing the identified plurality of data assets to be displayed on the computing device; receiving a first user selection of one or more of the identified plurality of data assets from the computing device; retrieving metadata of the one or more of the identified plurality of data assets from the catalog of data assets based on the first user selection; sending a first request to a large language model, the first request comprising the retrieved metadata and being configured to prompt the large language model to generate a plurality of data models based on the retrieved metadata; receiving the plurality of data models from the large language model; determining that one of the plurality of data models does not satisfy a rule, the rule comprising a format rule or a grammar rule; sending an indication that the one of the plurality of data models does not satisfy the rule to the large language model; receiving a corrected version of the one of the plurality of data models based on the sending of the indication; and causing the plurality of data models to be displayed on the computing device.
- 21.** The non-transitory machine-readable storage medium of claim 20, wherein the causing of the plurality of data models to be displayed on the computing device comprises rendering a visual representation of the corrected version of the one of the plurality of data models based on processing of code representing the corrected version of the one of the plurality of data models.
- 22.** The non-transitory machine-readable storage medium of claim 20, wherein one or more of the plurality of data assets comprises a table of data.
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